CALCULUS · LIMITS OF ALGEBRAIC FUNCTIONS

Limits Problem Set A

COPY 1 OF 2

NAME

CLASS

DATE

Evaluate each limit. Show every algebraic step — an unsupported answer scores nothing.

Time 40 min · Total 28 marks

1 Evaluate $\lim_{x \to 4} \dfrac{x^2 - 16}{x - 4}$

3 marks

2 Evaluate $\lim_{x \to 0} \dfrac{\sqrt{x + 16} - 4}{x}$

3 marks

3 Evaluate $\lim_{x \to 3} \dfrac{x^3 - 27}{x^2 - 9}$

3 marks

4 Evaluate $\lim_{x \to 2} \dfrac{x^3 - 8}{x^2 - 5x + 6}$

3 marks

5 Evaluate $\lim_{x \to \infty} \dfrac{3x^2 - 5x + 2}{4x^2 + 7x - 1}$ — 3 marks

6 Evaluate $\lim_{x \to \infty} \dfrac{\sqrt{4x^2 + 1}}{x + 3}$ — 4 marks

CHALLENGE SECTION

7 Evaluate $\lim_{x \to 1} \left(\dfrac{1}{1 - x} - \dfrac{3}{1 - x^3} \right)$ HARD — 4 marks

8 Evaluate $\lim_{x \to \infty} \left(\sqrt{x^2 + 3x} - \sqrt{x^2 - x} \right)$ HARD — 5 marks

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